

BUSA6004 – Introduction to Inference, Modelling, and Forecasting

Professional Practice: Business Inference, Modelling, and Forecasting Report

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I acknowledge that I have not used GAITs (e.g., ChatGPT) in drafting and proofreading this assignment.

Question 1 - Data Cleaning:

Assume that you are working as a business analyst for a property developer in Sydney. You are tasked with analysing a large dataset of property sales data. In the dataset, there is a column called Sort ID. Use the filter button to only select the Sort ID that matches the last digit of your student ID. The rest of this question should be based on this filter. You should be looking at just 300 rows of data. There are 5 cells which need to be cleaned. Explore your filtered data set to find the data that needs to be cleaned. Delete the entire row, if you find a cell that needs to be cleaned.

Answer

The 5 cells that need to be cleaned are presented in the following table:

Cell Content	Column
-2	Bedrooms
Agent Name	Agent Name
Cake	Property Type
2026	Year Built
0	Land Size (sqm)

Question 2 - Research on Lookup Functions:

There is a column called Sort ID. Use the filter button to only select the Sort ID that matches the last digit of your student ID. The rest of this question should be based on this filter. You should be looking at just 295 rows of data (since you deleted 5 rows in Question 1). Research how to use VLOOKUP and XLOOKUP. Using the sheet 'Council Area' and the functions: VLOOKUP and XLOOKUP, populate each of the columns Council Area 1 and Council Area 2 (in the main 'Data' sheet) – one for each Excel function. The answers should be the same across the two columns.

Answer

Formula used for the first cell of column Council Area 1:
 =VLOOKUP(B33, 'Council Area'!A2:B51, 2)

Annotation:

B33: The cell containing the value "Pymble" in row 33.

'Council Area'!A2:B51: The whole dataset of the “Council Area” sheet, except for the two heading cells of “Suburb” and “Council Area”.

Evidence from Excel:

	B	C
ID	Suburb	Council Area 1
1		
33	8 Pymble	Ku-ring-gai Council

Formula used for the first cell of column Council Area 2:

=XLOOKUP(B33, 'Council Area'!A2:A51, 'Council Area'!B2:B51)

Annotation:

B33: The cell containing the value “Pymble” in row 33.

'Council Area'!A2:A51: All the cells in the column “Suburb” of the “Council Area” sheet, excluding the heading cell.

'Council Area'!B2:B51: All the cells in the column “Council Area” of the “Council Area” sheet, excluding the heading cell.

Evidence from Excel:

	B	C	D
ID	Suburb	Council Area 1	Council Area 2
1			
33	8 Pymble	Ku-ring-gai Council	Ku-ring-gai Council

Question 3 - Data Summary:

There is a column called Sort ID. Use the filter button to only select the Sort ID that matches the last digit of your student ID. The rest of this question should be based on this filter. You should be looking at just 295 rows of data.

For the variable, Price:

- Provide numerical summaries of centred-ness and spread.

- Provide two graphical summaries where each of them showcases centred-ness and spread.

The two dot points below should be answered within 100-200 words:

- What are some insights that you can draw from your summaries above?
- What further steps would you take to provide more depth to these numerical and graphical summaries?

Answer

Numerical summaries of centred-ness for the variable *Price*:

- The mean:

The mean is calculated as follows in Excel:

D2		=AVERAGE(A2:A296)	
		C	D
1			
2		Mean	1,646,657.12

Result: 1,646,657.12

Annotation:

AVERAGE: The function to calculate the mean in a set of values

A2:A296: The array containing the data of 'Price' (excluding the 'Price' heading)

- The median:

The median is calculated as follows in Excel:

D3		=MEDIAN(A2:A296)	
		C	D
3		Median	1,609,946

Result: 1,609,946

Annotation:

MEDIAN: The function to calculate the median in a set of values

A2:A296: The array containing the data of 'Price' (excluding the 'Price' heading)

- The mode:

The mode is calculated as follows in Excel:

D4		: X ✓ fx		=MODE.MULT(A2:A296)	
		C		D	
4	Mode			#N/A	

Result: There is no mode

Annotation:

MODE.MULT: The function that returns the most frequent values in a set of values

A2:A296: The array containing the data of 'Price' (excluding the 'Price' heading)

Numerical summaries of spread for the variable *Price*:

- The range:

The range is calculated as follows in Excel:

D5		: X ✓ fx		=MAX(A2:A296)-MIN(A2:A296)	
		C		D	
5	Range			2,586,268	

Result: 2,586,268

Annotation:

MAX: The function that returns the largest value in a set of values

MIN: The function that returns the smallest value in a set of values

A2:A296: The array containing the data of 'Price' (excluding the 'Price' heading)

- The interquartile range

The interquartile range is calculated as follows in Excel:

D6	C	D
6	Interquartile Range	1,188,088

Result: 1,188,088

Annotation:

The interquartile range is the difference between the third quartile and the first quartile.

QUARTILE.INC(A2:A296, 3): The function to calculate the third quartile

QUARTILE.INC(A2:A296, 1): The function to calculate the first quartile

A2:A296: The array containing the data of 'Price' (excluding the 'Price' heading)

- The variance:

The variance is calculated as follows in Excel:

D7	C	D
7	Variance	524,117,227,167.46

Result: 524,117,227,167.46

Annotation:

VAR.S is the function to calculate the variance of a sample

A2:A296: The array containing the data of 'Price' (excluding the 'Price' heading)

- The standard deviation:

The standard deviation is calculated as follows in Excel:

D8	$\times$ $\checkmark$ f_x	=STDEV.S(A2:A296)
	C	D
8	Standard Deviation	723,959.41

Result: 723,959.41

Annotation:

STDEV.S: The function to calculate the standard deviation of a sample

A2:A296: The array containing the data of 'Price' (excluding the 'Price' heading)

- The coefficient of variation:

The coefficient of variation is calculated as the quotient of the standard deviation and the mean.

Result: 0.44

Two graphical summaries where each of them showcases centred-ness and spread for the variable *Price*:

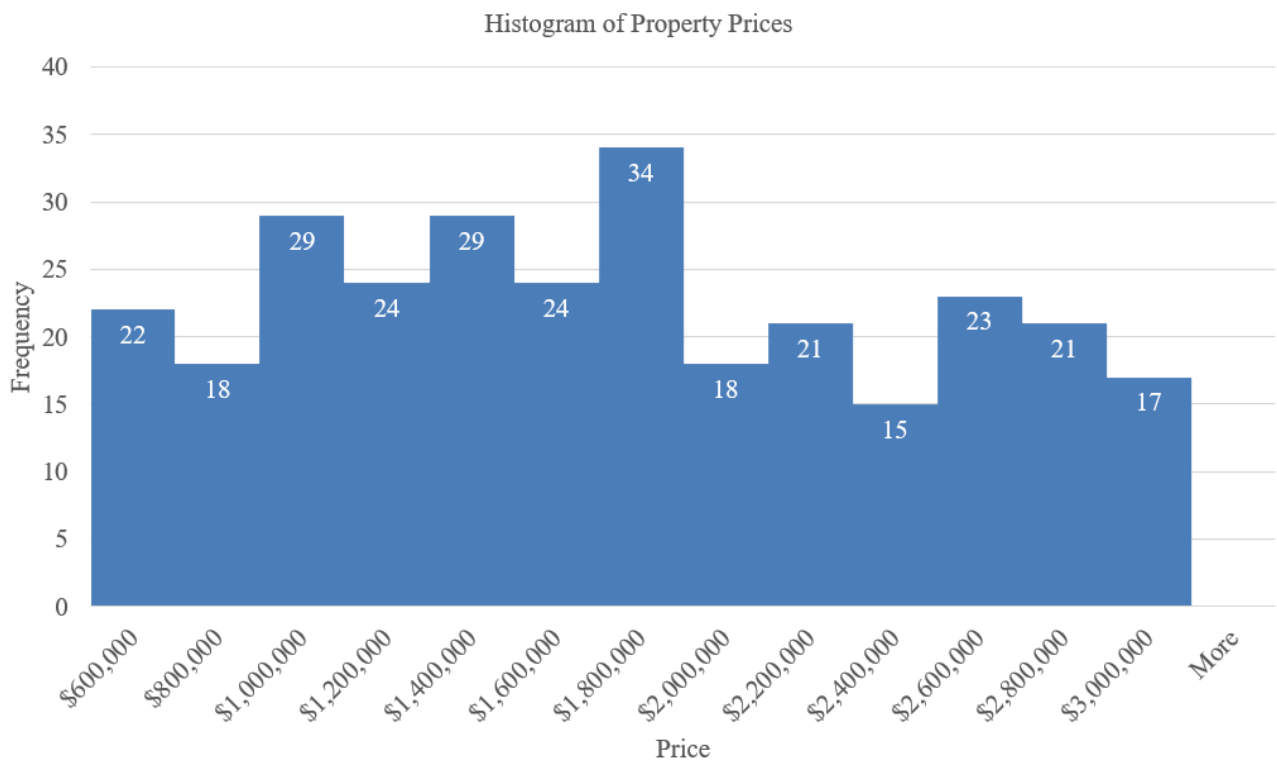
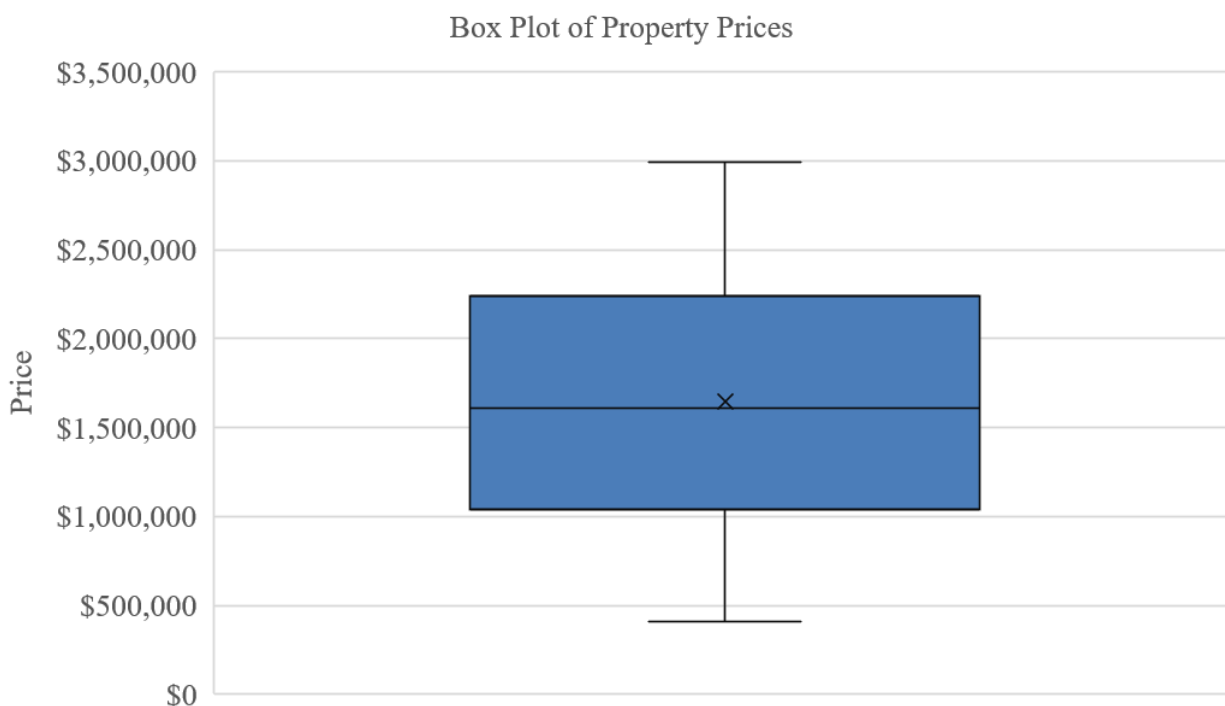


Figure 1. Histogram of Property Prices**Figure 2. Box Plot of Property Prices**

Several insights can be drawn from my summaries. The average property price (\$1,646,657.12) is somewhat greater than the median (\$1,609,946), indicating a modest right-skew in the distribution. This is reinforced by the histogram, which illustrates that most properties are concentrated in the lower-to-middle price categories, while a smaller number of higher-priced properties extend the upper range. The standard deviation (\$723,959.41) and variance (524,117,227,167.46) reflect a significant level of price variability. The interquartile range is \$1,188,088, indicating a sizable spread among the central 50% of values. The coefficient of variation (0.44) is also a substantial measure of relative variation. The box plot conveys property prices are spread from below \$500,000 to just under \$3,000,000.

To look deeper, I would break down those statistics by types of properties. T-tests would help discover if price differences among categories were statistically significant. Also, by comparing box plots or histograms for each type of property, differences in spread as well as spread skewness would appear. Furthermore, examining correlations for the additional factors (e.g., suburb or house area) may assist in providing explanatory factors for why certain types sell for more than others, or have greater variability in price.

Question 4 - Statistical Test

Turn the Sort ID filter off. You should be looking at 2995 rows of data.

Choose one of the following Property Types: Apartment, House, Townhouse, Villa (you will have to turn on another filter in your spreadsheet).

For the variable of your choosing:

- Perform a statistical test to test whether there is evidence that the average price of the Property Type of your choosing is greater than \$1.675 million.
- In your response, show all the steps of a hypothesis test.

The two dot points below should be answered within 100-200 words:

- What are some insights that you can draw from your test above?
- What further analysis would you conduct to provide more depth to this test?

Answer

H_0 : The average price of the villas is equal to \$1.675 million.

H_a : The average price of the villas is greater than \$1.675 million.

Sample size is large, so we use Z-statistics.

$$Z = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

$$\bar{x} \text{ (sample mean)} = 1,687,142.05$$

$$\mu_0 \text{ (hypothesised population mean)} = 1,675,000$$

$$s \text{ (sample standard deviation)} = 774,484.38$$

$$n \text{ (sample size)} = 725$$

$$Z = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{1,687,142.05 - 1,675,000}{774,484.38 / \sqrt{725}} = 0.42$$

The critical value at 5% using standard normal table (cumulative probability of 0.95) is 1.645.

Since Z-statistics (0.42) < 1.645, we do not reject H_0 . We conclude that there is not enough evidence to say that the average villa price is greater than \$1.675 million.

The findings suggest that the observed variation may be due to random sampling, not a real increase in average villa prices. To better understand villa prices, it would help to look at the data distribution more closely. This could include calculating and interpreting the 95% confidence interval for the mean to see the plausible range of true villa prices. Applying regression models could also help clarify which factors; such as land size, year built, or suburb; that are major drivers of villa prices. These insights assist sellers, buyers, and investors in grasping which property attributes have the most substantial impact on price beyond the mean, enabling decisions regarding renovations, marketing approaches, or investment opportunities to be more data-informed and efficient.

Question 5 - Correlation Analysis:

Now select all Property Types but turn the Sort ID filter on according to the last digit of your Student ID. You should be looking at 295 rows of data.

Examine the correlation between Bedrooms and House Area.

- Examine the graphical and numerical correlation.

The two dot points below should be answered within 100-200 words:

- What are some insights that you can draw from your analysis above?
- What further steps would you take to provide more depth to this correlation analysis?

Answer

H_0 : There is not a positive correlation between the number of bedrooms and house area.

H_a : There is a positive correlation between the number of bedrooms and house area.

Since we are looking at 295 rows of data, sample size $n = 295$

Correlation coefficient using Excel:

$$r = \text{CORREL}(\text{Bedrooms data array}, \text{House Area data array}) = 0.84$$

Evidence from Excel:

E2	=CORREL(A2:A296,B2:B296)	
1	D	E
2	Correlation coefficient	0.84

T-statistic:

$$t = \frac{r}{\sqrt{1-r^2}/\sqrt{n-2}} = \frac{0.84}{\sqrt{1-0.84^2}/\sqrt{295-2}} = 26.47$$

Critical value at 5% level using Excel:

$$t^* = T.INV(1 - \alpha, n - 2) = T.INV(1 - 0.05, 295 - 2) = 1.65$$

Evidence from Excel:

E4	=T.INV(1-0.05, 295-2)	
	D	E
4	Critical value at 5%	1.65

Since $|t| > t^*$, we reject H_0 . We conclude that there is a positive correlation between the number of bedrooms and house area.

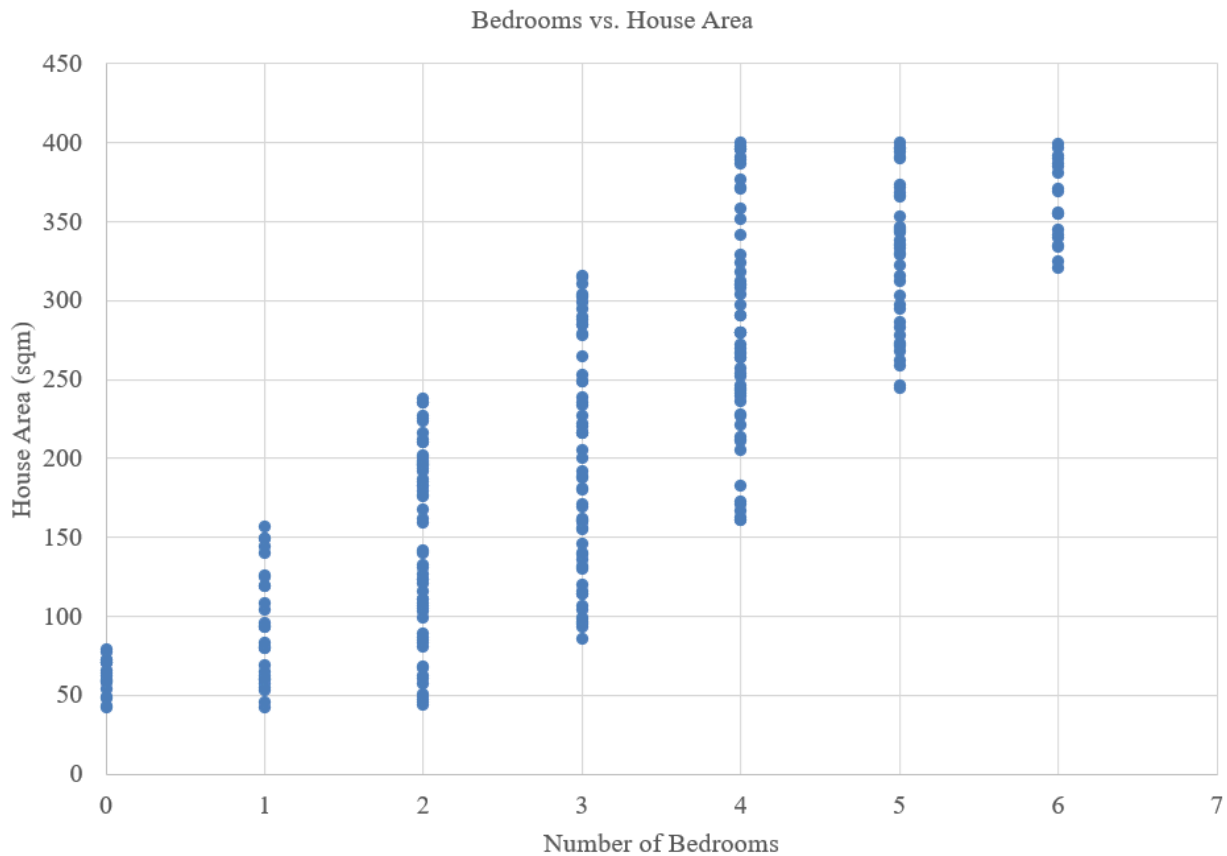


Figure 3. Scatter plot showing the correlation between Bedrooms and House Area

The analysis shows a strong connection ($r = 0.84$) between the number of bedrooms in a house and its overall size, which suggests that homes with more bedrooms typically have larger floor areas. This finding is statistically significant because the t value of 26.47 far exceeds the critical value of 1.65 at 5% significance level. The scatter plot also backs this up, clearly demonstrating that as the number of bedrooms increases, the size of the house tends to grow as well. This raises the possibility that the number of bedrooms could be a solid predictor of a home's total size, which could really help when it comes to pricing and planning for new developments.

To dig deeper, I would like to explore whether this relationship differs among various types of properties or suburbs. It would also be useful to factor in other elements, like land size and when the house was built, to see if the bedroom count is truly the main driver behind house size. A multiple regression analysis could give us a fuller picture of how these factors interact.

Question 6 - Regression:

Select all Property Types but turn the Sort ID filter on according to the last digit of your Student ID. You should be looking at 295 rows of data.

Construct a Regression Model based House Area, Bedrooms, and Bathrooms.

Include a screenshot of your regression results after you have professionally formatted your results in Excel.

Provide the full regression equation of the model by using the variable names above – use professional formatting of your equation.

Explain your coefficients, comment on the statistical significance of your regression results, and comment on the overall model.

Additionally, construct an appropriate Bubble Chart that represents all of the data. Make sure that you choose your axes carefully. Do not edit the minimum and maximum points of any axes. Include a screenshot of your Bubble Chart.

Answer

Screenshot of regression results:

Summary Output

Regression Statistics	
Multiple R	0.84
R Square	0.71
Adjusted R Square	0.70
Standard Error	58.48
Observations	295

ANOVA

	df	SS	MS	F	Significance F
Regression	2	2390220.5	1195110.25	349.4	0.00
Residual	292	998762.04	3420.42		
Total	294	3388982.54			

Coefficients

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	42.28	10.87	3.89	0.00	20.89	63.68	20.89	63.68
Bedrooms	56.63	2.15	26.34	0.00	52.40	60.86	52.40	60.86
Bathrooms	-1.74	4.28	-0.41	0.68	-10.15	6.68	-10.15	6.68

Regression equation:

$$\text{House Area} = 42.28 + 56.63 \times \text{Bedrooms} - 1.74 \times \text{Bathrooms}$$

Interpretation of coefficients:

- Intercept (42.28): If both bathrooms and bedrooms equal zero, the expected house area is 42.28 units.
- Bedrooms (56.63): With every extra bedroom, the house area increases by 56.63 units, provided that the number of bathrooms does not increase.
- Bathrooms (-1.74): Every extra bathroom decreases the house area by 1.74 units, keeping the number of bedrooms constant.

Statistical significance:

The number of bedrooms is a strong predictor of house area ($p < 0.001$). In contrast, the number of bathrooms does not hold statistical significance with $p = 0.68$, suggesting it does not meaningfully affect house area.

Comment on the model:

This regression model fits the data very well. With $R^2 = 0.71$, the model explains about 71% of the variation in House Area, which is very strong for real estate data. The number of bedrooms is a highly significant predictor of house area, while the number of bathrooms does not significantly influence house area once bedrooms are accounted for.

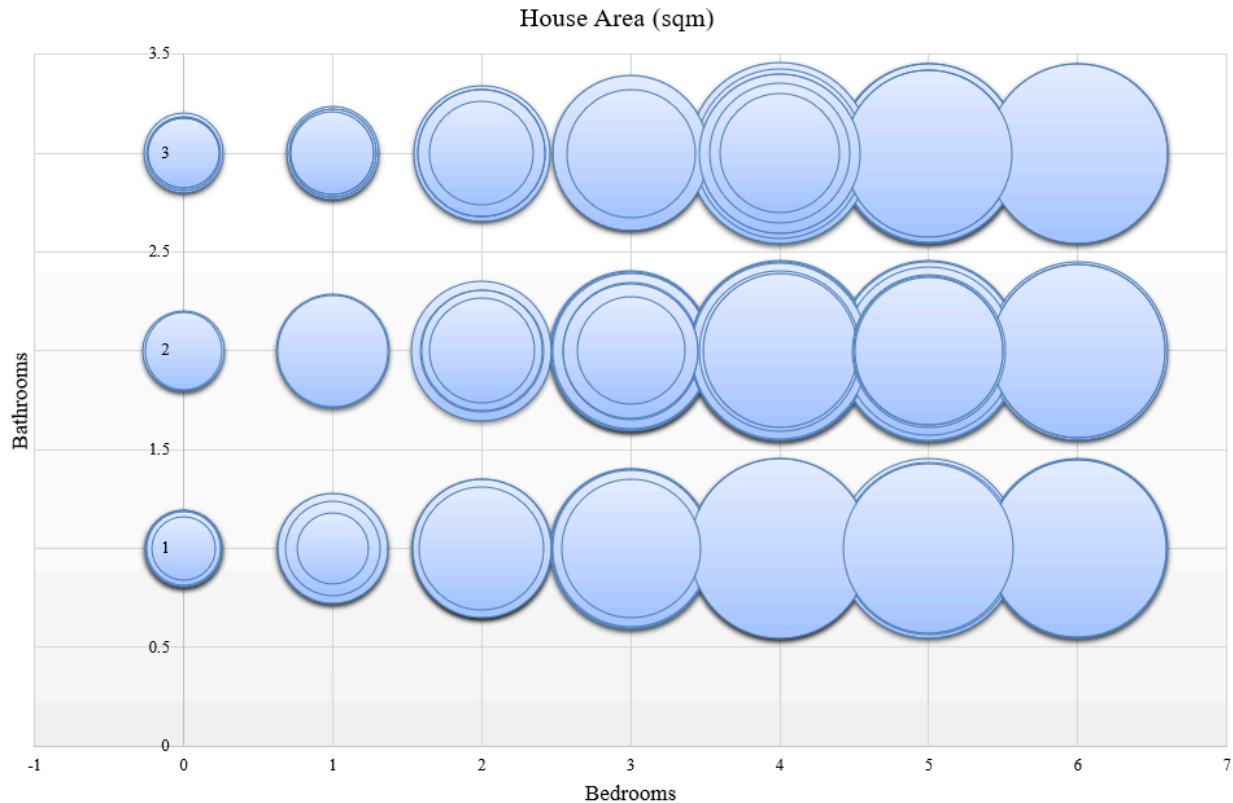


Figure 4. Bubble chart representing House Area, Bathrooms, and Bedrooms

Question 7 - Two-sample t-test:

Select all Property Types but turn the Sort ID filter on according to the last digit of your Student ID. You should be looking at 295 rows of data.

Research how to conduct a two-sample t-test in Excel.

Create a new sheet with 2 columns of data Apartment Prices and House Prices. You will have to filter your data to create these two new columns.

Conduct a two-sample t-test assuming unequal variances to test whether there is a significant difference between Apartment Prices and House Prices.

Include screenshots of your output (professionally formatted in Excel), explain what you did, and explain and interpret your findings.

Answer

Step 1: State the hypotheses

Null Hypothesis (H_0):

There is no difference in the mean prices of apartments and houses.

$$H_0: \mu_1 = \mu_2$$

Alternative Hypothesis (H_a):

There is a difference in the mean prices of apartments and houses.

$$H_a: \mu_1 \neq \mu_2$$

Step 2: Conduct a two-sample t-test assuming unequal variances in Excel:

- Filter the dataset to extract prices of apartments and houses.
- Put them in two separate columns in a new sheet.
- Enable the Data Analysis function in the Data tab.
- Choose t-Test: Two-Sample Assuming Unequal Variances and click OK.
- For Variable 1 Range: Select all Apartment Prices.
- For Variable 2 Range: Select all House Prices.
- Hypothesized Mean Difference: Enter 0.
- Check Labels if there are headings in the first row instead of data.
- Alpha: 0.05.
- Output Range: Choose the cell where the results appear and click OK.

The results are displayed below:

t-Test: Two-Sample Assuming Unequal Variances

	Apartment Price	House Price
Mean	1578217.93	1718257.07
Variance	531747690531.25	549738184666.53
Observations	69	74
Hypothesised Mean Difference	0	
df	141	
t Stat	-1.14	
P(T<=t) one-tail	0.13	
t Critical one-tail	1.66	
P(T<=t) two-tail	0.26	
t Critical two-tail	1.98	

The p-value (two-tailed) is 0.26, which is greater than 0.05, so we fail to reject H_0 and conclude that there is no difference in mean prices between apartments and houses. Houses certainly look to have a higher mean price, but the variability within each group is large enough that this difference may be attributable to random sampling.

Question 8 - ANOVA:

Select all Property Types but turn the Sort ID filter on according to the last digit of your Student ID. You should be looking at 295 rows of data.

Research how to conduct a Single Factor (One Way) ANOVA in Excel.

Create a new sheet with 4 columns of data: Apartment Prices, Townhouse Prices, House Prices, and Villa Prices. You will have to filter your data to create these four new columns.

Conduct a Single Factor (One Way) ANOVA to test whether there is a significant difference between the prices of these 4 categories.

Include screenshots of your output (professionally formatted in Excel), explain what you did, and explain and interpret your findings.

Answer

Step 1: State the hypotheses

Null Hypothesis (H_0):

The mean prices of all four property types are equal.

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

Alternative Hypothesis (H_a):

At least one mean price is different from the others.

$$H_a: \text{At least one } \mu \text{ differs}$$

Step 2: Conduct a Single Factor (One Way) ANOVA in Excel

- Filter the dataset to extract prices of apartments, townhouses, houses, and villas into a new sheet.
- Put them in four separate columns in a new sheet.
- Enable the Data Analysis function in the Data tab.
- Choose Anova: Single Factor and click OK.
- For Input Range, select the whole new sheet just created.
- For Grouped by, choose Columns.
- Check “Labels in first row” if there are headings in the first row instead of data.
- Alpha: 0.05.
- Output Range: Choose the cell where the results appear and click OK.

The results are displayed below:

Summary

Groups	Count	Sum	Average	Variance
Apartment Prices	69	108897037	1578217.93	531747690531.25
Townhouse Prices	79	129167881	1635036.47	503272132484.77
House Prices	74	127151023	1718257.07	549738184666.53
Villa Prices	73	120547910	1651341.23	525426147639.65

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	714825386585.59	3	238275128861.87	0.45	0.72	2.64
Within Groups	153375639400648	291	527064052923.19			
Total	154090464787234	294				

The p-value is 0.72, which is higher than 0.05. Therefore, we cannot reject H_0 and conclude that there is no difference in the mean prices of apartments, townhouses, houses, and villas. The variation in prices is likely explained by other factors than property type alone.

Question 9 - In-class question:

a) Consider Question 1. (200 – 400 words)

- i. In the working life of a Business Analytics Professional, briefly explain why completing such a task is useful in fulfilling their role.
- ii. Briefly describe how you performed the tasks in Question 1.
- iii. When answering Question 1. What were the most challenging parts of the task? What were the easiest parts of the task?

b) Consider Question 2. (200 – 400 words)

- i. In the working life of a Business Analytics Professional, briefly explain why completing such a task is useful in fulfilling their role.
- ii. Briefly describe how you performed the tasks in Question 2.
- iii. When answering Question 2. What were the most challenging parts of the task? What were the easiest parts of the task?

c) Consider Question 3. (200 – 400 words)

- i. In the working life of a Business Analytics Professional, briefly explain why completing such a task is useful in fulfilling their role.
- ii. Briefly describe how you performed the tasks in Question 3.
- iii. When answering Question 3. What were the most challenging parts of the task? What were the easiest parts of the task?

Answer

a) In the working life of a Business Analytics Professional, data cleaning is essential. It makes sure the data used for analysis is accurate. Removing errors and duplicates helps provide reliable insights and prevents misleading conclusions. For this question, I began with adjusting the Sort ID column by using the filter button to only select the Sort ID that matches the last digit of my student ID, which is 8. Then I continued to use the Sort ID button on other columns to spot anomalies. In the column Bedrooms, I spotted a value of '-2' which is unreasonable as the number of bedrooms should not be negative. In the column Agent Name, I spotted the value

‘Agent Name’ which could be attributed to mistakes during input. In the column Property Type, I spotted the value ‘Cake’, which is a type of dessert. In the column Year Built, I spotted the value ‘2026’, which is unreasonable again as the year is still futuristic. In the column Land Size (sqm), I spotted the value ‘0’ meaning the house does not occupy any area of land. The final step was to delete the 5 rows containing these errors. To me, the most challenging part for this question was knowing how to use the filter buttons. I tried to look for anomalies using my eyes at first but looking at 300 rows of data could be very confusing. The easiest part was obtaining the results after managing to use the filter buttons as the anomalies are immediately revealed.

b) In the working life of a Business Analytics Professional, using VLOOKUP and XLOOKUP functions allows them to quickly find and match data in large datasets, which saves time and cuts down on manual errors. For this question, I researched how to use the two functions online and found the main difference was XLOOKUP has the ability to search for values in both columns to the left and right of the lookup array, whereas VLOOKUP can only find values located to the right of the first column in the specified array. I started with the VLOOKUP function for the first cell of Council Area 1 column, for which I inputted the cell containing the value “Pymble” in row 33 and the whole dataset of the “Council Area” sheet, except for the two heading cells of “Suburb” and “Council Area”. Then I proceeded to the XLOOKUP function for the first cell of Council Area 2 column, for which I inputted three things: the cell containing the value “Pymble” in row 33; all the cells in the column “Suburb” of the “Council Area” sheet, excluding the heading cell; and all the cells in the column “Council Area” of the “Council Area” sheet, excluding the heading cell. The final step was to obtaining the results, formally formatting them before copying them to the report. The most challenging part for me in this question was to research the two functions and execute them as I have never used them before. I had to juggle between different Google Search results to find the one that is most accurate and updated, then I had to follow it step-by-step and check if it was done correctly. The easiest part was to professionally format the results by adjusting the font of the results and taking the screenshots.

c) In the working life of a Business Analytics Professional, providing summaries of centred-ness and spread not only reveal patterns about business performance or customer behavior, help analysts spot unusual or incorrect data but also act as simplified metrics that enable managers to make informed and data-driven decisions.

The numerical summaries of centred-ness for the variable *Price* in the dataset are the mean, median, and mode. The numerical summaries of spread for the variable *Price* in the dataset are the range, interquartile range, variance, standard deviation, and coefficient of variation. I used the functions provided in Microsoft Excel to calculate these values, obtained the results, professionally formatted them and copied them to the report.

For the mean, I used the function =AVERAGE(data array).

For the median, I used the function =MEDIAN(data array).

For the mode, I used the function =MODE.MULT(data array).

For the range, we know that it is the difference between the maximum value and the minimum value of the data, so I used the function MAX(data array) - MIN(data array).

For the interquartile range, we know that it is the difference between the third quartile and the first quartile of the data, so I used the function QUARTILE.INC(data array, 3) - QUARTILE.INC(data array, 1).

For the variance, I used the function =VAR.S(data array).

For the standard deviation, I used the function =STDEV.S(data array).

For the coefficient of variation, I calculated the quotient of the standard deviation and the mean.

The graphical summary of centred-ness for the variable *Price* in the dataset is the histogram, while that of spread is the box plot.

To generate the histogram, I created a new column of “Bins” detailing 13 different groups of price, ranging from \$600,000 to \$3,000,000 based on the minimum and maximum values. Then I used the Data Analysis ToolPak of Microsoft Excel to create the histogram by inputting the range of the data array and the newly created bins. To generate the box plot, I selected the data array containing the values for the variable *Price*, then went to the *Insert* tab, then *Insert Statistic Chart*, then inserted the *Box and Whisker* chart. The last step is always to professionally format the visuals before copying it to the report.

After that, I proceeded to write the summary of insights drawn from the summary as well as suggest further steps to provide more depth for the analysis. In my opinion, this was the most difficult part to deal with as it requires critical thinking as well as the ability to interpret the results. In contrast, the most straightforward part was to obtain the summaries as the instructions to do that on Microsoft Excel are already provided during my lectures and seminars.